

Non-Hermitian Absorbing Layers for Schrödinger's Smoke

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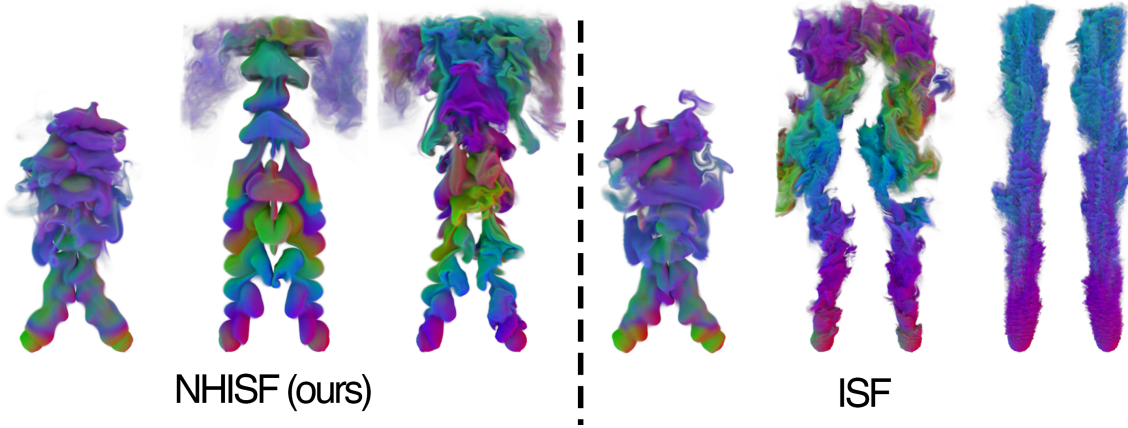


Figure 1: Two columns of buoyant smoke. When ISF is solved with periodic boundary conditions, accelerated velocity circulates from top to bottom, making the velocity undesirably high (right). By introducing our non-Hermitian Hamiltonian absorbing layer, velocity is damped at the ceiling, allowing the two smoke columns to merge and settle to a steady state(left).

ABSTRACT

This paper proposes novel open boundary conditions for incompressible Schrödinger flow (ISF) by introducing a new non-Hermitian term to the original Schrödinger's equation, effective only within narrow-banded layers. Unlike previous work that explicitly requires auxiliary variables, our method achieves the same objective without such complexity, facilitating implementation. We demonstrate that our method retains the benefits of the original ISF while robustly dissipating velocity without noticeable reflections.

CCS CONCEPTS

• Computing methodologies → Physical simulation.

KEYWORDS

open boundary condition, incompressible Schrödinger's flow

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1 INTRODUCTION

Incompressible Schrödinger flow (ISF), proposed by Chern et al [Chern et al. 2016], is a promising method for simulating incompressible fluids while preserving coherent vortical structures by using complex wave functions. However, it is limited to periodic or wall-bounded domains, leading to unphysical behavior when simulating an infinitely extended area. To overcome this problem, open boundary conditions based on the perfectly matched layer (PML) method have been proposed in the field of fluid simulation [Chern 2019; Söderström et al. 2010; Söderström and Museth 2009]. However, these previous studies cannot be applied naively because the primary physical quantity of the ISF is the wave function rather than the velocity. Furthermore, the PML method often requires auxiliary variables, increasing memory consumption and computational complexity.

In this paper, we propose a novel extension to the ISF that incorporates open boundary conditions without explicit auxiliary variables. Our approach introduces a non-Hermitian absorbing layer near the domain boundaries, effectively dissipating velocity and enabling the simulation of open boundaries while retaining the key advantages of the original ISF formulation.

In summary, our non-Hermitian term for realizing open boundary conditions is succinct without auxiliary variables, unconditionally stable, and a highly efficient algorithm with additional costs only within the narrow-band absorbing layer.

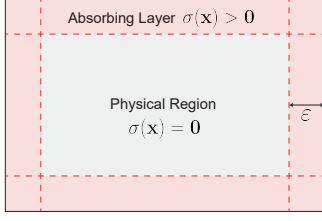


Figure 2: Our grid structure. We lay absorbing layers (shown in pink) at the edges of the domain boundary with each having the thickness of ϵ to realize open boundary conditions.

2 METHOD

Our non-Hermitian ISF (NHISF) introduces a dissipative non-Hermitian Hamiltonian term $-i\hat{H}_{\text{damp}}$ to the original Schrödinger’s equation

$$i\hbar \frac{\partial}{\partial t} \psi(\mathbf{x}, t) = (\hat{H}_{\text{ISF}} - i\hat{H}_{\text{damp}}) \psi(\mathbf{x}, t), \quad (1)$$

where $\psi(\mathbf{x}, t) \in \mathbb{C}^2$ and $\hbar \in \mathbb{R}$ denote two-component wave function and a parameter which characterizes the quantization of circulation. $\hat{H}_{\text{ISF}}$ and $-i\hat{H}_{\text{damp}}$ denote a Hamiltonian operator introduced in ISF and our new non-Hermitian term, which enables dissipation of the velocity. More specifically, they are given by

$$\hat{H}_{\text{ISF}} := -\frac{\hbar^2}{2} \nabla^2 + p(\mathbf{x}, t), \quad (2)$$

$$\hat{H}_{\text{damp}} := -\nabla \cdot \text{diag}(\sigma_x(\mathbf{x}), \sigma_y(\mathbf{x}), \sigma_z(\mathbf{x})) \nabla. \quad (3)$$

The term $-i\hat{H}_{\text{damp}}$ is designed to be non-zero only within narrow absorbing layers placed near the domain boundaries, as defined by a strength function $\sigma(x)$. It controls the absorption strength independently along each axial direction (Figure 2). We found that the smooth step function proposed in Söderström et al. [Söderström et al. 2010] as a $\sigma(x)$ provides effective dissipation and minimal reflection.

The non-Hermitian term $-i\hat{H}_{\text{damp}}$ is discretized using a standard 7-point Laplacian stencil on a uniform grid and integrated by an implicit Euler method before pressure projection.

Due to its localization within the narrow absorbing layers, it can be efficiently solved without modifying the entire domain. We provide a proof demonstrating that the non-Hermitian term acts as a velocity damper that exponentially decays fluid velocity within the absorbing layers (see supplemental material).

3 RESULTS AND DISCUSSION

The NHISF method is evaluated on two benchmark scenarios: a buoyant smoke scene (Figure 1) and a smoke jet scene (Figure 3).

In the buoyant smoke scene, periodic boundary conditions lead to unnatural acceleration due to the velocity circulating from the top to the bottom of the grid. In contrast, our method allows the absorbing layer on the ceiling to dissipate the velocity, enabling the two rising smoke columns to reach a steady state (Figure 1).

In the jet example, the absorbing layers on the right side dissipate the outward velocity. In contrast, periodic boundaries result in unphysical recirculation of the fluid velocity, which can be observed from the motion of the tracer particles (Figure 3).

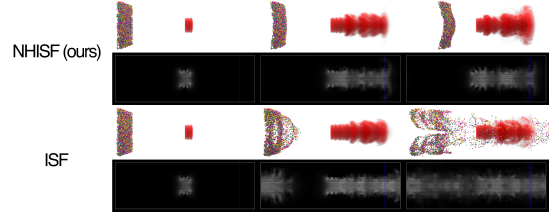


Figure 3: Jet. Our method is shown in the top two rows, each showing both rendered 3D and the sliced 2D velocity view. Corresponding simulation using ISF [Chern et al. 2016] with periodic boundary conditions is shown in the bottom two rows. Our method dissipates the velocity at the right edge of the grid, preventing unphysical velocity recirculation.

Our non-Hermitian term is integrated using the implicit Euler method, making it unconditionally stable. In fact, we were able to achieve stable simulations using a time step size of $\Delta t = 1/24$ in both examples. This is in contrast to PML methods, which are subject to CFL number restrictions [Söderström and Museth 2009].

The total computation time per frame in the buoyant smoke example is 23.54 seconds, of which 24.4% is spent on calculating the non-Hermitian term. In the jet smoke example, the total computation time per frame is 3.27 seconds, with 44.0% spent on the non-Hermitian term. Our non-Hermitian Hamiltonian term can be solved only for the degrees of freedom in the narrow-banded region, so the velocity dissipation can be computed in the same or less computation time as the remaining terms that are computed rapidly using FFT.

4 CONCLUSIONS AND FUTURE WORK

We have demonstrated that by adding a narrow-banded dissipative non-Hermitian term to ISF, unconditionally stable open boundary conditions can be realized without explicit auxiliary variables.

However, NHISF has limitations, such as the possibility of reflections due to discretization error and the lack of systematic parameter selection. Despite these limitations, we believe our method opens up further exploration of dissipative quantum systems and their application to fluid simulation in computer graphics.

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